Paolo Ruffini — Teoria generale delle equazioni (1799)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Paolo Ruffini of Modena (1765–1822), physician and mathematician, who made the first sustained attempt to prove that the great door does not open — his proof carried a gap, specified after Rosen (1995), and Abel completed it in 1824 — and was left standing before it unheard.

The author's own terms

Function, in his sense (§1): “any analytical expression whatever in which the variable, or the variables, are contained alone or joined with constant quantities.” By permutation (§§2–3): *f(x, y, z)* keeps its value under every permutation; *f(x)(z)* changes; mixed groupings such as *f(x, y)(z)* are unchanged under some, changed under others. Functions from one expression are similar; from different expressions, dissimilar (§4). One acquiring never more than one value is uniform; more, multiform — biform, triform, quadriform by count (§6). His general determinate algebraic equation (§17) yields every particular equation by fixing exponent and coefficients.

The story

Paolo Ruffini was born at Valentano in September 1765, son of the physician Basilio Ruffini; the family removed to Modena, where he spent his life. At its university he studied medicine, philosophy, literature, and mathematics. In 1788, at twenty-three, he took degrees in philosophy, medicine, and mathematics, became professor of the foundations of analysis, and in 1791 took the elementary mathematics chair and a medical licence.

Appointed against his wishes to the Cisalpine Republic's Junior Council in 1796, he refused the oath without the qualification his conscience required, and was dismissed; the Austrian return in 1799 restored him. That year the *Teoria* appeared. Lagrange never replied; the Paris Academy would not take the memoir up. He re-argued it in 1803, 1806, and 1813, against Malfatti and Fontana. He declined a call to Pavia rather than leave his patients; in 1814 he became rector of Modena's re-established university. Cauchy, persuaded by studying him, wrote in 1821 that his memoir had proved the insolubility conclusively — with this dossier's qualification: his proof carried a gap, specified after Rosen (1995), and Abel completed it in 1824. Ruffini died at Modena on 10 May 1822. He had caught typhus in 1817, recovered, and never regained his strength; whether it directly caused his death is not certified, and is not asserted here.

The work and what survives

Teoria generale delle equazioni, in cui si dimostra impossibile la soluzione algebraica delle equazioni generali di grado superiore al quarto (Bologna: Stamperia di S. Tommaso d'Aquino, 1799; base text ETH-Bibliothek Zürich, e-rara, DOI 10.3931/e-rara-15207) runs to twenty chapters on his own map: Chapter 1, functions; Chapters 2, 3, 9, 10, properties; Chapters 4–8, transformations; Chapters 11–12, the third and fourth degrees; Chapter 13, the impossibility demonstration; Chapters 14–16 and the end of 20, reducible equations; Chapters 17–20, approximation.

This project holds an opening fascicle only, in progress: the Preliminary Discourse (pp. III–VIII) and Chapters 1–2 in draft translation. Chapter 1 (§§1–16) is entire. Chapter 2 reaches §32, at a joint Ruffini himself makes, but unevenly: within §§17–32, §§25–29 are partly bracketed editorial summaries, not full translation — §25 breaks off into a bracketed note, §26 is a bracketed

summary, and §§27–29 a bracketed summary. It is not a uniform full translation, and nothing in it is his complete proof (Chapter 13, unreached). Beyond it survive the remaining chapters in Italian, his re-arguments, and the *Opere matematiche* (Bortolotti's edition).

Its English fortunes

No English translation of this work, whole or in substantial part, was found after research.

English exposition of the proof exists in quantity; it is not a translation. The theorem is now taught as the Abel–Ruffini theorem: Ruffini made an incomplete proof in 1799, refined in 1813 — his proof carried a gap, specified after Rosen (1995) as assuming the radicals in a supposed solution lay, by field operations alone, in the splitting field of the roots — and Abel completed the proof in 1824.

What completing it would take

First, complete §§25–29 at full length, and carry §§33 onward — the power-sums and Newtonian theorems. Then the book entire: Chapters 3–20, eighteen chapters, including Chapter 13. No completion claim may be made before Chapter 13 is reached, and the gap must be presented there as here, named at once. The whole must pass the folder's gate: review by a reader competent in modern algebra before any claim of readiness, and the editor's word before anything is published or shelved.

Sources

- Project folder: `~/workspace/rare-and-obscure-books/books/ruffini-teoria-generale/ruffini-teoria-generale.txt` (base copy: ETH e-rara, DOI 10.3931/e-rara-15207)
- PIN: `~/workspace/packages/boneyard-extension/research/PIN-ruffini.md`
- MacTutor, Ruffini: https://mathshistory.st-andrews.ac.uk/Biographies/Ruffini/
- *Catholic Encyclopedia* (1912), Ruffini: http://studylight.org/

encyclopedias/eng/tce/p/paolo-ruffini.html

- MAA *Convergence*, Ruffini's Theory of Equations: https://old.maa.org/press/periodicals/convergence/mathematical-treasure-ruffini-s-theory-of-equations
- Oxford lecture record: https://courses.maths.ox.ac.uk/mod/resource/view.php?id=3093